Lesson 14
Concentration Inequalities

A concentration inequality gives us a probability band on certain random variables taking atypically large or atypically small values (tail probabilities)

$\mathbb{P}(X > a)$ $a \in \mathbb{R}^+$

$\mathbb{P}(X < -a)$

1. Markov's Inequality

Assume X is a non-negative r.v. with $\mathbb{E}[X] < \infty$, then for any $\alpha > 0$

$$\mathbb{P}(X > \alpha) \leq \frac{\mathbb{E}[X]}{\alpha}$$

Remark: This inequality is non-trivial when $\alpha > \mathbb{E}[X]$.

large deviations

$X > \alpha$

$B = \{X > \alpha\}$ is an event

$I_B(X)$ is a Bernoulli r.v.

$I_{X>\alpha}(X)$ is a Bernoulli r.v.

$\forall x \quad I_B(x) + I_{B^c}(x) = 1$

$I_{X>\alpha}(X) + I_{X\leq\alpha}(X) \underset{\sim}{=} 1$ w.p.1.

$E[X] = E[X(1)] = E\left[X I_{X>\alpha}(X) + X I_{X\leq\alpha}(X)\right]$

$= E\left[X I_{X>\alpha}(X)\right] + \underbrace{E\left[X I_{X\leq\alpha}(X)\right]}_{\geq 0}$

$\Rightarrow E[X] \geq E[\underbrace{X I_{X>\alpha}(X)}]$

$$X I_{X>\alpha}(X) = \begin{cases} 0 & X \leq \alpha \\ X & X > \alpha \end{cases}$$

$$\mathbb{E}[X] \geq \mathbb{E}[X I_{X>\alpha}(X)]$$

$$\geq \mathbb{E}[\alpha I_{X>\alpha}(X)]$$

$$= \alpha \underbrace{\mathbb{E}[I_{X>\alpha}(X)]}_{\mathbb{P}(X>\alpha}$$

$$\underbrace{\mathbb{P}(X>\alpha)} \leq \frac{E[X]}{\underbrace{\alpha}}$$

As an exercise, prove Markov ineq. for continuous r.v.'s using integrals & def. of expectation.

Question: The "decay rate" is $1/\alpha$ in Markov Inequality.

Can we have a decay rate that is faster than $1/\alpha$?

The answer is YES:

2. Chebyshev's Inequality

Assume that X is a r.v. with expectation μ and variance $\sigma^2 < \infty$, then

$$\mathbb{P}(|X-\mu| > k\sigma) \leq 1/k^2$$

$$\forall k > 0$$

OR

$$\mathbb{P}(|X-\mu| > c) \leq \frac{\sigma^2}{c^2}$$

$k\sigma = c$ $\quad \mathbb{P}(|X-\mu| > c) \leq \frac{1}{\left(\frac{c^2}{\sigma^2}\right)} = \frac{\sigma^2}{c^2}$

$k = \frac{c}{\sigma}$

Proof:

$$\mathbb{P}(|X-\mu| > c) = \mathbb{P}(|X-\mu|^2 > c^2)$$

$$\leq \frac{\mathbb{E}[|X-\mu|^2]}{c^2} = \frac{\sigma^2}{c^2}$$

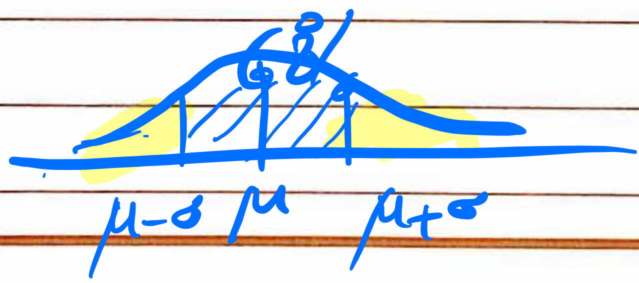
By Markov inequality

Put $c = k\sigma$ to get the other version.

Chebyshev's Inequality for Various k's

$k=1 \quad \mathbb{P}(|X-\mu| > \sigma) \leqslant \frac{1}{1^2}$

when X is Normal $\approx 32\%$

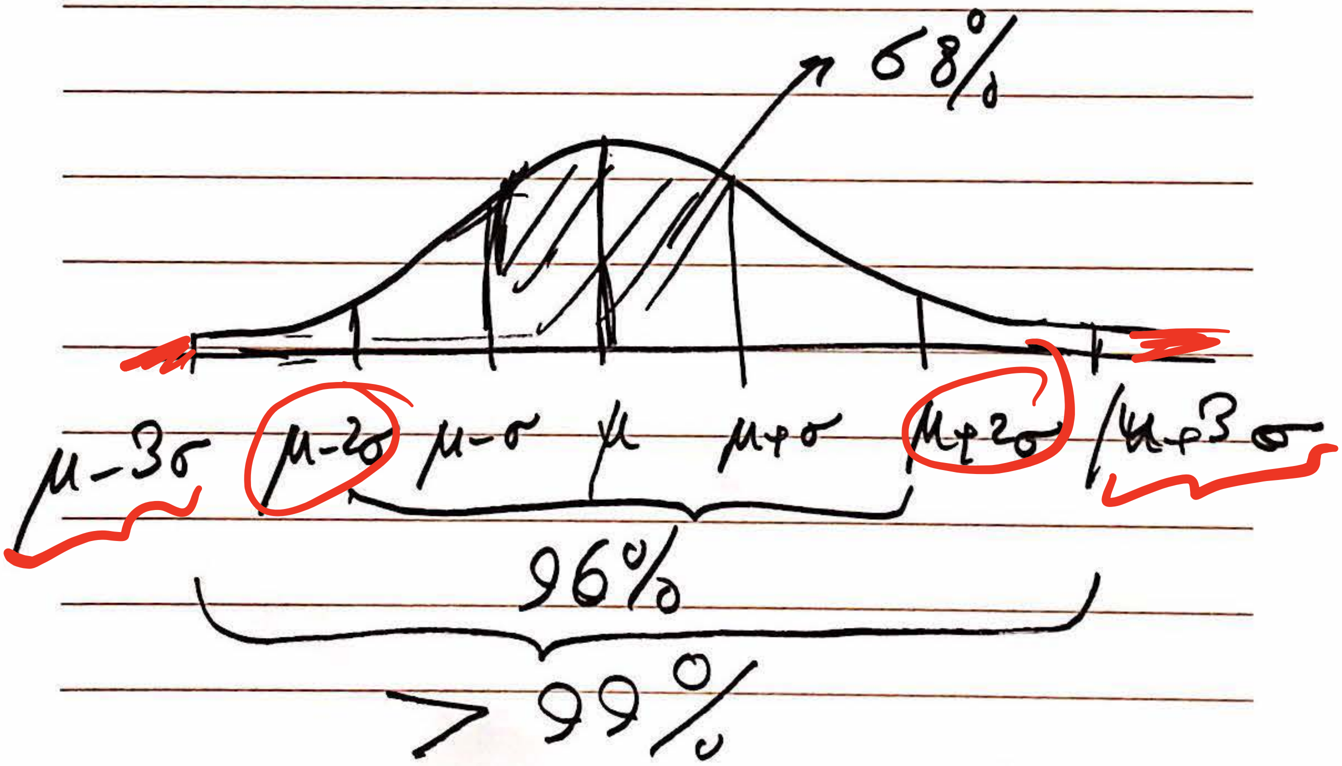


$k=2$ $\quad \mathbb{P}(|X-\mu|>2\sigma) \leq \frac{1}{2^2}$

For Normal $\approx 4\%$

$k=3$ $\quad \mathbb{P}(|X-\mu|>3\sigma) \leq \frac{1}{9}$

For normal $<1\%$



Chebyshev Inequality yields a bound that decays as fast as $1/k^2$, an improvement over the basic Markov Inequality.

Question: Can we find bounds that decay faster than $1/$polynomias, i.e., exponentially?

YES! Chernoff Bound